```
planetary/
├── aura_core/                  # [LOCKED] Immutable Math Engine
│   ├── equations/
│   │   ├── KTE_equation.py       # Kardashev Transition
│   │   ├── CEI_equation.py       # Civilization Evolution
│   │   ├── SKI_equation.py       # Sustainable Kardashev
│   │   ├── transition_rate.py    # R_t = T_i + A_i + P_i
│   │   ├── time_to_type1.py      # TT1 logic
│   │   ├── co_evo_ledger.py      # Tab 4 Resource/Debt Ledger
│   │   ├── transition_diff.py    # dK/dt logistic growth
│   │   └── coherence_eq.py       # Coherence C logic
│   ├── twins/                  # Digital Twin simulations
│   │   ├── Earth1_historical/    # Conservative baseline
│   │   └── Earth2_accelerated/   # AI-optimized path
│   ├── constants.py              # Physics/Civilizational constants
│   └── __init__.py               # Core API
├── uncertainty_engine/           # Monte Carlo & Confidence Intervals
├── causality_engine/             # Attribution & Change Analysis
├── governance/                   # Ethical Guards & FORB-Registry
├── macro_simulation/             # Long-horizon S-curves & Shocks
├── collective_intelligence/      # Coordination & Sentiment (C-input)
└── data_validation/              # Input integrity stubs
```